To combine the observations of the preceding two paragraphs, imagine a sequence of snapshots of a Calabi-Yau space in which the size of a particular three-dimensional sphere gets smaller and smaller. The first observation implies that a three-brane wrapping around this three-dimensional sphere—which appears to us as a black hole—will have ever smaller mass until, at the final point of collapse, it will be massless. But, as we asked above, what does this mean? The answer became clear to us by invoking the second observation. Our work showed that the new massless pattern of string vibration arising from the space-tearing conifold transition is the microscopic description of a massless particle into which the black hole has transmuted. We concluded that as a Calabi-Yau shape goes through a space-tearing conifold transition, an initially massive black hole becomes ever lighter until it is massless and then it transmutes into a massless particle—such as a massless photon—which in string theory is nothing but a single string executing a particular vibrational pattern. In this way, for the first time, string theory explicitly establishes a direct, concrete, and quantitatively unassailable connection between black holes and elementary particles.

“Melting” Black Holes

The connection between black holes and elementary particles which we found is closely akin to something we are all familiar with from day-to-day life, known technically as a phase transition. A simple example of a phase transition is the one we mentioned in the last chapter: water can exist as a solid (ice), as a liquid (liquid water), and a gas (steam). These are known as the phases of water, and the transformation from one form to another is called a phase transition. Morrison, Strominger, and I showed that there is a tight mathematical and physical analogy between such phase transitions and the space-tearing conifold transitions from one Calabi-Yau shape to another. Again, just as someone who has never before encountered liquid water or solid ice would not immediately recognize that they are two phases of the same underlying substance, physicists had not realized previously that the kinds of black holes we were studying and elementary particles are actually two phases of the same underlying stringy material. Whereas the surrounding temperature determines the phase in which water will exist, the topological form—the shape—of